Energy and Environmental Economics
ECON 3381/5137, ENVR 4931/5331 Spring 2024
University of Houston – Clear Lake

Course Information

Instructor: Allyssa A. Wadsworth, PhD, Assistant Professor of Economics
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Office Location: Bayou Building Suite 3121 #33

Class Meets: Wednesday 7:00pm – 9:50pm, BB2524
Office Hours: Tuesday 4pm – 6pm, Thursday 10am – 12pm, & by appointment.
Office Hours Location: Remote via Zoom (link available on Canvas) and in-person
Course Type: Face-to-face
Credit Hours: 3
Prerequisites: Principles of Macroeconomics (ECON2301), Principles of Microeconomics (ECON2302)

Required Text

Students are required to purchase both of the following resources:

- Dahl, Carol A., "International Energy Markets: Understanding Pricing, Policies, and Profits," 2nd ed., 2015, PennWell, ISBN 978-1-59370-291-5
- Tietenberg, Tom and Lynne Lewis, "Environmental Economics and Policy," 7th ed., 2020, Routledge, ISBN 9780429995118
- Supplemental Material: Students will be assigned additional readings and videos throughout the class to supplement and enhance the material in the textbook.

Course Objectives

Undergraduates: This is a survey course in energy and environmental economics, with a few topics in natural resource economics. Each of these can be a standalone semester course but we will be allocating the class to all three topics.

Upon successful completion of the course, students will be able to:

- Explain the basic tenets of environmental and natural resource economics
- Understand the basic structure of US energy markets
- Critically analyze and evaluate the types, reasons for, and effectiveness of government intervention in resource and energy markets
- Students will exit the course with a strong foundation for subsequent instruction in applied microeconomics.

The course has three exams, a presentation, two written assignments, and several homework assignments. Undergraduates will receive a multiple-choice final exam.

Graduates: You cover all of the above, but are expected to show a greater depth of understanding than your undergraduate counterparts.

Upon successful completion of the course, students will be able to:

- Complete all of the requirements above
- Synthesize information to construct policy recommendations
- Understand and utilize policy reports from the US government, the Energy Information Administration, and the private sector's BP Statistical Energy Review.

The course has three exams, a final paper, and several homework assignments. Graduate students will receive a multiple-choice and short answer final exam.

Grading Policies

Undergraduates:

Activity	Percent of Grade
Presentation (1)	10%
Homework (7)	30%
Real World Response (2)	20% (10% each)
Exams (Best 2 out of 3)	40% (20% each)

Graduates:

Activity	Percent of Grade
Homework (7)	30%
Exams (Best 2 out of 3)	50% (25% each)
Final Paper (1)	20%

Grading Scale: Letter grades will be given using the traditional scale described in the table below.

Letter Grade	Percentage
A	$93 \leq$ Score
A-	$90 \leq$ Score < 93
B+	$87 \leq$ Score < 90
B	$83 \leq$ Score < 87
B-	$80 \leq$ Score < 83
C+	$77 \leq$ Score < 80
C	$73 \leq$ Score < 77
C-	$70 \leq$ Score < 73
D+	$67 \leq$ Score < 70
D	$63 \leq$ Score < 67
D-	$60 \leq$ Score < 63
F	Score < 60

Grade Availability and Appeals:
Grades will be posted to Canvas no later than two weeks (14 days) after an assignment is turned in or an exam is completed. Students who believe there is a discrepancy in one of their grades should email the instructor as soon as possible and describe their concern in detail. The appeal will be reviewed accordingly and grades adjusted if appropriate. Students will receive an email reply to their grade appeal within the period of time described below in "Class Policies" section which will include an explanation of the grade adjustment or instructions for further review of the course materials.

Incomplete Grade:
An incomplete may be awarded to a student who can prove in a single submission of verifiable documentation that they suffered an unexpected catastrophic event after the withdrawal date and had a passing grade in the course at the time. This must be in accordance to the university guidelines. All assignments and exams must be taken as soon as possible to fill in the missing grades.

Assignments

Students are strongly encouraged to complete all assignments through Canvas, where all of the course materials and instructions are posted. Homework and other graded assignments are all completed and submitted online. Students are expected to complete all assigned homework and other assignments on time.

- Late submissions of Homework will have an automatic percent reduction of a flat 25% and may be turned in at any time until the last scheduled class day. This policy excludes presentations, papers, and exams. No assignments will be accepted during exam week.

Homework
Undergraduate: Multiple choice, drawing, and open ended questions based on knowing basic facts, being able to solve easy to moderate calculation problems, graphical analysis and drawing graphs. These assignments will be completed in Canvas and submitted by the due dates listed below. Late homework will receive -25%.

Graduate: Multiple choice and open-ended questions the require the ability to synthesize information and issue policy recommendations, graphical analysis and drawing graphs. Calculations will be moderate to difficult. These assignments will be completed in Canvas and submitted by the due dates listed below. Late homework will receive -25%.

Presentation (Undergraduate)
The presentations will be only completed by undergraduate students. You will be assigned one (1) peer-reviewed published paper on a topic we are coving below to present. The presentation will be no longer than 15 minutes with 5 minutes for audience questions. You will turn in your PowerPoint. More information will be provided as well as a rubric. If you miss the day you are scheduled to present you must do a video recording of your presentation for submission to receive credit.

Real World Response (Undergraduate)
Two written assignments worth 10% each. These assignments are short essays evaluating what is happening now on the topics we cover. Locate a news article about an Environmental or Energy

topic we have covered in class. Write an essay on the importance of the event and how it incorporates economics. Each essay should be 2-pages, double-spaced, Times New Roman, 12-point font, 1-inch margins. If you use AI in any way (to source your article, write the essay, etc.), please indicate so and explain the benefits of using, the drawbacks, **and** double check the factual information provided by the AI system (note: not all of the articles AI provides are real and not all of the factual information provided is accurate). More information will be provided.

Final Paper (Graduate)
This final paper will be only completed by graduate students. This paper is a 8-12 page paper, double spaced, 12-point font, Times New Roman, 1-inch margins, with references, and not written by an AI software (Trust me, we can tell.). Using the Annual Energy Outlook and the BP Statistical Energy Review, as well as other credible/peer reviewed sources, answer one of the following questions:

1. Look at the Annual Energy Outlook for 2013 as well as the current year. How has the development of electric vehicles changed the forecast for future energy consumption. What do you think energy consumption will look like by 2035 across both the transportation fuel and electricity sectors, and why. Detail any political assumptions you are making regarding policies in the future, as well as any realistic assumptions about the progress of technology over this time period. Include, by 2035, your estimate of what percentage of the US fleet and new vehicle sales (these are different numbers) will be electric.

2. Using the Annual Energy Outlook and BP Statistical Energy Review, what is the feasibility of the United States meeting its share of climate change goals. Consider what the appropriate goal should be (+1.5 degrees Celsius over preindustrial averages by 2100? +2.0 degrees by 2100? By 2200?) and then detail what policies should be enacted in order to achieve them. You should acknowledge both the benefits and the drawbacks of your proposed set of policies while arguing, of course, that your idea is the best path forward.

Rubric	**Percent of Grade**
Grammar	15%
Style/Flow (No fluff or time wasting)	25%
Avoids mistakes/logical errors. Uses correctly cited facts to support arguments.	30%
Clear progression of paper (Thesis – Supporting Arguments – Conclusion)	30%

Exam Policy

There will be three (3) exams given in accordance to the class schedule below. Exams will be held in person. They will be no more than 90-minutes long and given during class and finals week from 7:00pm until 8:30pm. Each exam is worth 20% (25% graduate) of your final grade and will divide up the chapters. A review sheet will be provided 1-week prior to the exam. All lecture material, readings from the text, supplemental material, and homework questions are potential sources for exam questions.

There will be no make-up exams. Instead, only two (2) out of the three (3) exams will be counted for your final grade. If you miss one (1) exam the other two (2) exams will be counted. If you miss two (2) exams, one (1) of the missed exams will be counted as a zero (0).

Cheating and Academic Integrity:
Students are expected to follow the University's statement on academic integrity in "UHCL Academic Honestly Policy." All exams and graded assignments will be monitored closely for cheating, plagiarism (including of other students' work), and other academic dishonesty. Students caught cheating will be reported to the appropriate University authorities.

GroupMe: You are encouraged to utilize a GroupMe (or similar conversation app) for this class. I use a common textbook that has many of the answers online for the assignments. Therefore, it is okay to work together on the assignments and utilize all of the tools available. Please note that if you solely look up the answers and not learn how/why, you will find the exams harder in this class.

While utilizing GroupMe or any other communication method, if you share information about an exam with other students, you have committed an academic honesty violation. The first person to report someone to me who is subsequently convicted of doing so receives 10 points on their course grade. This incentive is designed to prevent cheating, but it also works as enforcement.

Extra Credit

I add 2 percentage points to your final score at the end of the semester. This is a buffer for when life unexpectedly gets in the way and you are late on assignments or may not do as well on a submission as you hoped. In exchange for this coverage, you are expected to avoid doing any of the following:

1. Request or demand to be accommodated/extended on homework assignments for any reason. These assignments are open for numerous weeks.
2. Request or demand extra credit, both in the form of more work or just a bump in grade.
3. Request or demand a calculation of your grade before the end of semester. The formula and Excel grade calculator is given to you.
4. Schedule an appointment with me and not attend the appointment. If you can not attend, please just notify me.
5. Request or demand to reschedule an exam or widen the available exam window. The dates are in the syllabus right now and the provisions for missing an exam are already provided.

If you do any of the above, you will forfeit the 2 percentage points of extra credit and will likely be told no anyway.

Class Policies

Email:

- Before sending email, students should read all instructional material posted to Canvas to find answers to questions that have already been provided. Announcements on Canvas will be sent as email messages.
- When sending email, students may expect the following email response times:
    - Messages sent during the business week (M-F, 9am-5pm), should receive a reply within 24 hours.
    - Messages sent during weekends should receive a reply by the end of the workday on Monday.
    - I will reply to emails only during the business week (M-F, 9am-5pm).
- Email messages should never include confidential information such as grades, personal identifying information (PII), or other sensitive material.
- Emails should be written in a professional style with proper greeting and grammar.
- **Include ECON 3381, ECON 5137, ENVR 4931, or ENVR 5331 in the title of the email. (This helps me sort through my inbox and find your email easier.)**

Classroom and Canvas: *My classroom is a safe space for learning and discussion.*

- Ask questions about the material in class. If you have a question, someone else will most likely have the same question. I love questions! They are fun to work through as a class, shows that you are engaged, and provides feedback on what needs to be discussed in more depth.
- Attendance is not taken for this class but attending is important since this is a lecture-based class.
- Be polite in class and on Canvas discussions.
- If an appointment is made to meet with me, please show up. If you cannot make your appointment, email me in advance.
- I have an open-door policy. If you see my office door open feel free to stop in! 😊

Course Schedule Outline (Subject to Change)

Notes: Submission due date times are 11:59pm for all assignments. All exams will be completed during regular class time and the university scheduled final exam time.

D = Dahl (International Energy Markets)
LT = Lewis and Tietenberg (Environmental Economics and Policy)

	Dates	Chapters Covered and Exams	Assignments and Due Dates
1	1/15 – 1/21	Syllabus and course structure Introduction: LT Chapter 1 LT Chapter 2	Due: Wednesday 1/31 Homework #1 (LT1, LT2, review of Microeconomics)
2	1/22 – 1/28	Nonmarket Valuation: LT Chapter 3 LT Chapter 4	
3	1/29 – 2/4	Nonmarket Valuation: LT Chapter 5 Externalities and Transportation: D Chapter 4 D Chapter 11	Due: Wednesday 2/14 Homework #2 (LT3, LT4, LT5, D4, D11, LT12)
4	2/5 – 2/11	Externalities and Transportation: LT Chapter 12	
5	2/12 – 2/18	**Exam #1 Wednesday 2/14** Exam start time: 7pm Chapters: LT1, LT2, LT3, LT4, LT5, D4, D11, LT12	
6	2/19 – 2/25	Climate Change: LT Chapter 8 LT Chapter 9	Due: Wednesday 3/6 Homework #3 (LT8, LT9, LT10, LT11)
7	2/26 – 3/3	Climate Change: LT Chapter 10 LT Chapter 11	
8	3/4 – 3/10	Natural Resource Economics: LT Chapter 13 D Chapter 14	Due: Wednesday 3/20 Homework #4 (LT12, D14) Real World Response #1

9	3/11 – 3/17	Spring Break – No Classes	
10	3/18 – 3/24	Coal and Gas: D Chapter 3 D Chapter 4 (review)	Due: Wednesday 4/3 Homework #5 (D3, D4, D8)
11	3/25 – 3/31	Coal and Gas: D Chapter 8	
12	4/1 – 4/7	**Exam #2 Wednesday 4/3** Exam start time: 7pm Chapters: LT8, LT9, LT10, LT11, LT12, D14, D3, D4, D8	
13	4/8 – 4/14	Oil and Transportation: D Chapter 7 D Chapter 14 (review)	Due: Wednesday 4/17** Homework #6 (D7, D14)
14	4/15 – 4/21	Electricity Markets: D Chapter 5 D Chapter 6	Due: Wednesday 5/1 Homework #7 (D5, D6, D15) Real World Response #2
15	4/22 – 4/28	Electricity Markets: D Chapter 6 D Chapter 15	
16	4/29 – 5/6	**Exam #3 Wednesday 5/1** Exam start time: 7pm Chapters: D7, D14, D5, D6, D15	

Standard University Policies

Statement on Disability:

The University of Houston System complies with Section 504 of the Rehabilitation Act of 1973 and the Americans with Disabilities Act (ADA) of 1990, pertaining to the provision of reasonable academic adjustments/auxiliary aids for students with a disability. In accordance with Section 504 and the ADA guidelines, each university within the system strives to provide reasonable academic adjustments/auxiliary aids to students who request and require them. If you believe that you have a disability requiring an academic adjustment/auxiliary aid, please contact the **Accessibility Support Center** at https://www.uhcl.edu/accessibility-support-center/ or disability@uhcl.edu or 281-283-2648. If you have an accommodation, please let me know the first day of class.

Statement of Academic Honesty:

The Academic Honesty Policy at UHCL (found on the Dean of Students' website, the Faculty Handbook, the Student Handbook, the Senior Vice President and Provost's website, the Graduate Catalog, and the Undergraduate Catalog) states: Academic honesty is the cornerstone of the

academic integrity of the university. It is the foundation upon which the student builds personal integrity and establishes a standard of personal behavior. Because honesty and integrity are such important factors in the professional community, you should be aware that failure to perform within the bounds of these ethical standards is sufficient grounds to receive a grade of "F" in this course and be recommended for suspension from UHCL. The Honesty Code of UHCL states "I will be honest in all my academic activities and will not tolerate dishonesty."

Withdrawal Information
Please be aware of the final day to drop the class in the academic calendar. Please also be aware of the 6 Drop Rule Limitation. Students who entered college for the first time in Fall 2007 or later should be aware of the course drop limitation imposed by the Texas Legislature. Dropping this or any other course between the first day of class and the census date for the semester/session does not affect your 6 drop rule count. Dropping a course between the census date and the last day to drop a class for the semester/session will count as one of your 6 permitted drops. You should take this into consideration before dropping this or any other course. Visit www.uhcl.edu/records for more information on the 6 drop rule and the census date information for the semester/session.

Statement on Assessment
The University of Houston-Clear Lake may use educational assessment tools in this course. “Educational Assessment” is defined as the systematic collection, interpretation, and use of information about student demographics, educational environments, learning outcomes, and professional success. These assessment tools will not affect student grades, but will provide faculty with confidential detailed information that will be used to improve student learning, courses, curriculum, and program accreditation.

Syllabus Changes
The instructor may need to make modifications to the course syllabus and may do so at any time. Notice of such changes will be announced as quickly as possible through the announcements on Canvas.

Recording of Class
Students may not record all or part of class, livestream all or part of class, or make/distribute screen captures, without advanced written consent of the instructor. If you have or think you may have a disability such that you need to record class-related activities, please contact the Accessibility Support Center. If you have an accommodation to record class-related activities, those recordings may not be shared with any other student, whether in this course or not, or with any other person or on any other platform. Classes may be recorded by the instructor. Students may use instructor’s recordings for their own studying and notetaking. Instructor’s recordings are not authorized to be shared with anyone without the prior written approval of the instructor. Failure to comply with requirements regarding recordings will result in a disciplinary referral to the Dean of Students Office and may result in disciplinary action.